

CHAPTER 9

overview of volatility modeling



The aim of this Chapter...

... is to give a hint towards the possible ways to model volatility, the most important parameter in derivatives pricing.

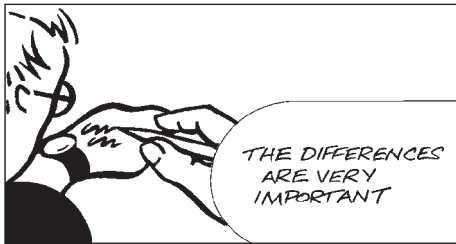
In this Chapter...

- a look at the joys of volatility modeling

9.1 INTRODUCTION

In the main, the Black–Scholes model is very robust and does a decent job of pricing derivatives, including exotics. One of the most important flaws in the model concerns the behavior of volatility. Quite frankly, we do not know what volatility currently is, never mind what it may be in the future. And the correct pricing of derivatives requires us to know what the volatility is going to be.

For this reason, volatility analysis and modeling takes a prominent role in the working life of a quant.



9.2 THE DIFFERENT TYPES OF VOLATILITY

We can't see volatility in the same way we can see stock prices or interest rates. The best we can hope to do is to measure it statistically. But such a measure is necessarily backwards looking, and we really want to know what volatility is going to be in the future.¹

For this reason people talk about different volatilities, as proxies for the real thing. Here are a few adjectives you can use with 'volatility.'

- Actual
- Historical/realized
- Implied
- Forward

9.2.1 Actual volatility

This is the measure of the amount of randomness in an asset return at any particular time. It is very difficult to measure, but is supposed to be an input into all option pricing models. In particular, the actual (or 'local') volatility goes into the Black–Scholes *equation*.

- There is no 'timescale' associated with actual volatility, it is a quantity that exists at each instant, possibly varying from moment to moment.

Example: The actual volatility is now 20% ... now it is 22% ... now it is 24% ...

9.2.2 Historical or realized volatility

This is a measure of the amount of randomness over some period in the past. The period is always specified, and so is the mathematical method for its calculation. Sometimes this backward-looking measure is used as an estimate for what volatility will be in the future.

- There are two 'timescales' associated with historical or realized volatility: one short and one long.

¹ You can see I am already assuming that volatility is not the nice constant that earlier chapters may have led you to believe.

Example: The 60-day volatility using daily returns. Perhaps of interest if you are pricing a 60-day option, which you are hedging daily.

In pricing an option we are making an estimate of what actual volatility will be over the lifetime of the option. After the option has expired we can go back and calculate what the volatility actually was over the life of the option. This is the realized volatility.

Example: I sold a 30-day option for a 30% volatility, I hedged it every day. Did I make money?

9.2.3 Implied volatility

The implied volatility is the volatility which when input into the Black–Scholes option pricing formulæ gives the market price of the option. It is often described as the market’s view of the future actual volatility over the lifetime of the particular option.

However, it is also influenced by other effects such as supply and demand.

- There is one ‘timescale’ associated with implied volatility: expiration.

Example: A stock is at 100, a call has strike 100, expiration in one year, interest rate 5% and the option market price is \$10.45. What volatility are traders using?

9.2.4 Forward volatility

The adjective ‘forward’ can be applied to many forms of volatility, and refers to the volatility (whether actual or implied) over some period in the future.

- Forward volatility is associated with either a time period, or a future instant.

9.3 VOLATILITY ESTIMATION BY STATISTICAL MEANS

9.3.1 The simplest volatility estimate: constant volatility/moving window

If we believe that volatility is constant or slowly varying, and we have N days’ data, we can use

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N R_i^2$$

where

$$R_i = \frac{S_i - S_{i-1}}{S_{i-1}}$$

and is the return on the i th day. Note that I haven’t annualized quantities yet.

There are obvious major problems associated with this volatility measure, because the returns are equally weighted you will get a plateauing effect associated with a large return. If there is a large one-day return it will increase the volatility instantaneously, but the estimate of volatility will stay raised until N days later when that return drops out of the sample. This is a totally spurious effect. This effect can be seen in Figure 9.1.

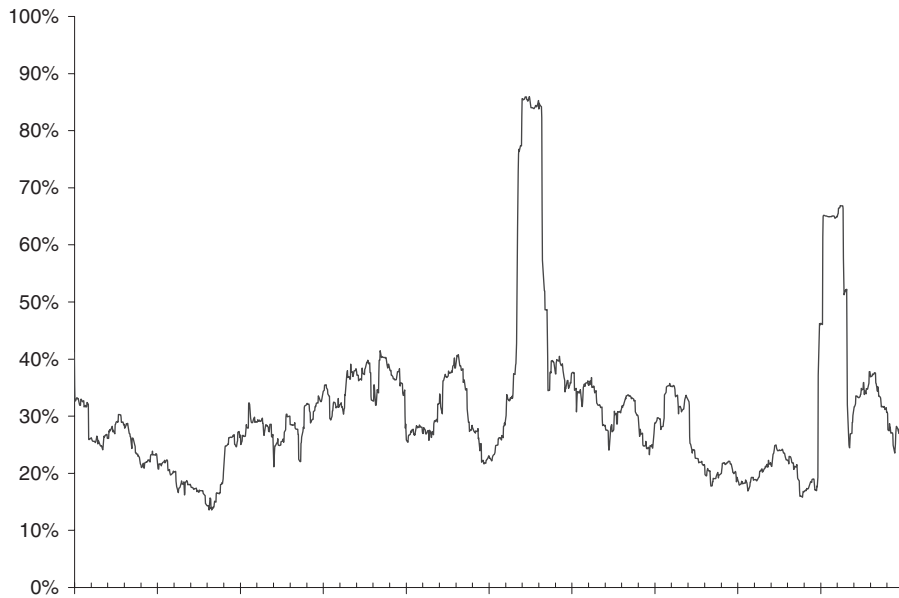


Figure 9.1 Moving-window volatility, observe the plateauing.

9.3.2 Incorporating mean reversion

Now let's consider time-varying volatility. We don't just have one σ but must consider σ_n , our estimate of the volatility on the n th day, using data available up to that point. If we believe that volatility tends to vary about a long-term mean $\bar{\sigma}$, then we could use

$$\sigma_n^2 = \alpha \bar{\sigma}^2 + (1 - \alpha) \frac{1}{n} \sum_{i=1}^n R_i^2.$$

Here there is a weighting assigned to each long-run volatility estimate and the current estimate based on the last n returns.

This is called an **ARCH** model, for **Autoregressive Conditional Heteroscedasticity**. But why should each of the last n returns be equally important?

9.3.3 Exponentially weighted moving average

Consider this estimate for volatility:

$$\sigma_n^2 = (1 - \lambda) \sum_{i=1}^{\infty} \lambda^{i-1} R_{n-i+1}^2.$$

The parameter λ must be greater than zero and less than one. This is an example of an **exponentially weighted moving average estimate**. The more recent the return, the more weight is attached. The sum extends back to the beginning of time.

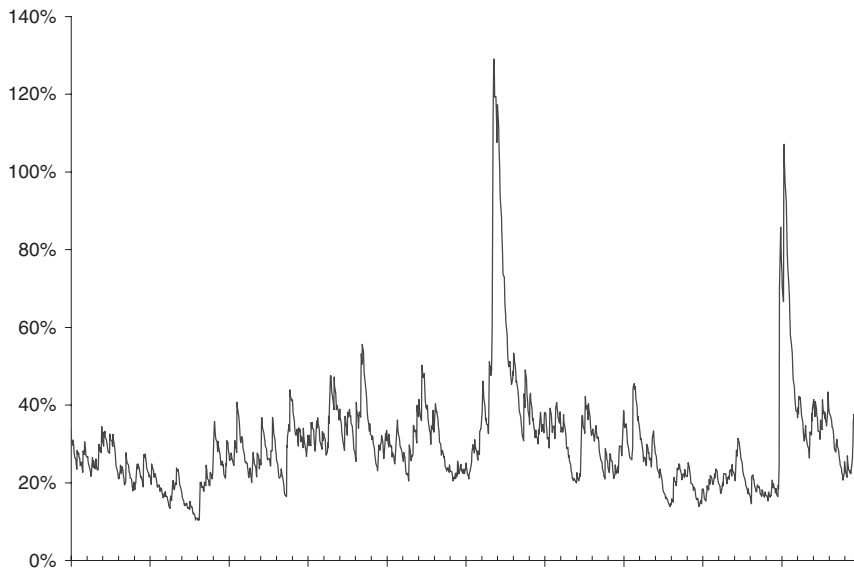


Figure 9.2 Exponentially weighted volatility, no plateauing.

The coefficient of $1 - \lambda$ ensures that the weights all add up to one. This expression can be simplified to

$$\sigma_n^2 = \lambda \sigma_{n-1}^2 + (1 - \lambda) R_n^2.$$

Note that this uses the most recent return *and* the previous estimate of the volatility. *This is RiskMetrics volatility measure.*

Figure 9.2 uses the same stock price data as in Figure 9.1 but now there is no plateauing.²

9.3.4 A simple GARCH model

Put the preceding models together to get

$$\sigma_n^2 = \alpha \bar{\sigma}^2 + (1 - \alpha) \left(\lambda \sigma_{n-1}^2 + (1 - \lambda) R_n^2 \right).$$

This is a **Generalized Autoregressive Conditional Heteroscedasticity** or **GARCH** model.

9.3.5 Expected future volatility

We are currently at day n and we want to estimate the volatility k days into the future, i.e. on day $n + k$.

² You could say that the obvious exponential decay in the volatility estimate is just as spurious as the discontinuity in the equally weighted case.

Expected future volatility: EWMA

Recall

$$\sigma_{n+k}^2 = \lambda \sigma_{n+k-1}^2 + (1 - \lambda) R_{n+k}^2.$$

Take expectations of this ...

$$E[\sigma_{n+k}^2] = \lambda E[\sigma_{n+k-1}^2] + (1 - \lambda) E[R_{n+k}^2].$$

But, of course,

$$E[R_{n+k}^2] = \sigma_{n+k}^2.$$

Therefore

$$E[\sigma_{n+k}^2] = \lambda E[\sigma_{n+k-1}^2] + (1 - \lambda) E[\sigma_{n+k}^2]$$

or, on rearranging,

$$E[\sigma_{n+k}^2] = E[\sigma_{n+k-1}^2].$$

In other words, the expected future value of the variance is the same as the previous day's value. Working backwards to the present, the expected future variance is the same as today's.

Expected future volatility: GARCH

Recall

$$\sigma_{n+k}^2 = \alpha \bar{\sigma}^2 + (1 - \alpha) (\lambda \sigma_{n+k-1}^2 + (1 - \lambda) R_{n+k}^2).$$

Taking expectations of this results in

$$E[\sigma_{n+k}^2] = \alpha \bar{\sigma}^2 + (1 - \alpha) (\lambda E[\sigma_{n+k-1}^2] + (1 - \lambda) E[R_{n+k}^2]).$$

On rearranging, we get

$$E[\sigma_{n+k}^2] = \frac{\alpha \bar{\sigma}^2}{1 - (1 - \alpha)(1 - \lambda)} + \frac{\lambda(1 - \alpha)}{1 - (1 - \alpha)(1 - \lambda)} E[\sigma_{n+k-1}^2].$$

The next expected value of the variance is a weighting between the most recent value and the long-term mean. Looking further into the future

$$E[\sigma_{n+k}^2] = \bar{\sigma}^2 + (E[\sigma_n^2] - \bar{\sigma}^2) (1 - \nu)^k$$

where

$$\nu = \frac{\alpha}{1 - (1 - \alpha)(1 - \lambda)}.$$

The path of expected future volatility³ is shown in Figure 9.3.

³ Actually variance, of course, which is anyway more important than volatility.

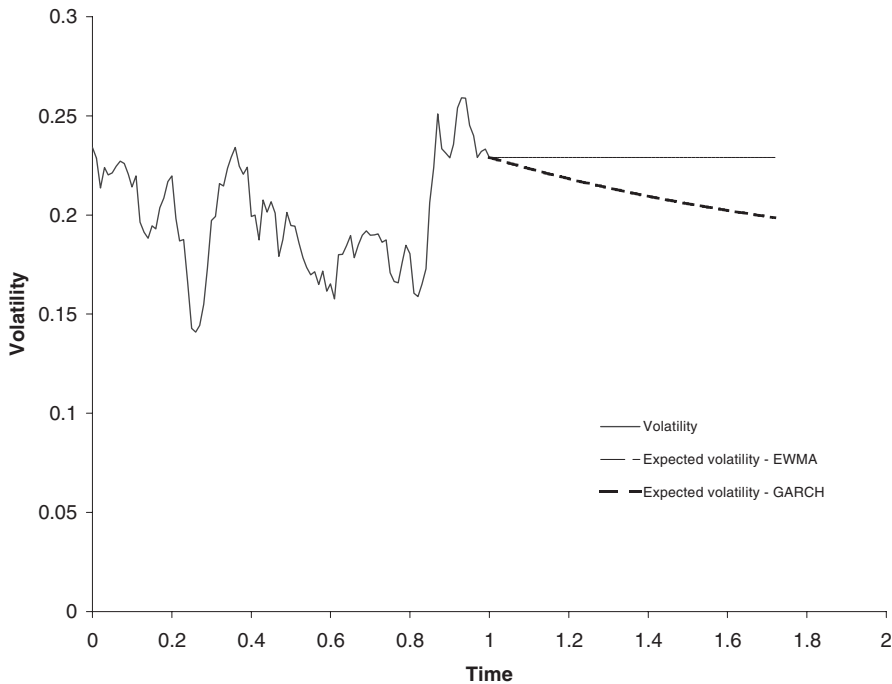


Figure 9.3 Expected future volatility for EWMA and GARCH models.

9.3.6 Beyond close-close estimators: range-based estimation of volatility

The problem with estimating volatility is that you need lots and lots of data to avoid sampling-error problems.

But then if you use too many days' worth of data you will be trying to estimate a parameter during a period when that parameter is almost certainly varying.

A catch-22 situation.

Or is it?

Why not use more information contained within a single day? That is, go down to finer timescales for the data. The problem with that is the behavior of returns over very short timescales, such as minutes, does not appear to be Normally distributed... there is even some evidence that the returns do not have a finite standard deviation.

Setting aside such worries(!), let's look at very simple ways of better estimating volatility using readily available price data, and not just closing prices.

Traditional close-to-close measure

When drift is small

$$\sigma_{cc}^2 = \frac{1}{n} \sum_{i=1}^n \left(\log \left(\frac{C_i}{C_{i-1}} \right) \right)^2.$$

Here there is a slight change of notation from before; C_i is the closing price on the i th day. Note also that we are looking at logarithms.

To adjust this for the drift take

$$\sigma_{acc}^2 = \frac{1}{n-1} \sum_{i=1}^n \left(\left(\log \left(\frac{C_i}{C_{i-1}} \right) \right)^2 - \frac{\left(\log \left(\frac{C_n}{C_0} \right) \right)^2}{n(n-1)} \right).$$

(Don't forget to annualize: multiply by the square root of the number of trading days in a year.)

Parkinson (1980)

This estimator uses extreme value, the highs H and the lows L during the day.

$$\sigma_p^2 = \frac{1}{4n \log(2)} \sum_{i=1}^n \left(\log \left(\frac{H_i}{L_i} \right) \right)^2.$$

This is five times more efficient than the close-to-close estimate. (That means, for the same amount of data the variance of the data is one fifth that of the close-to-close measure.)

Garman & Klass (1980)

At 7.4 times more efficient than close to close, we have

$$\begin{aligned} \sigma_{gk}^2 = \frac{1}{n} \sum_{i=1}^n & \left(0.511 \left(\log \left(\frac{H_i}{L_i} \right) \right)^2 \right. \\ & \left. - 0.019 \log \left(\frac{C_i}{O_i} \right) \log \left(\frac{H_i L_i}{O_i^2} \right) - 2 \log \left(\frac{H_i}{O_i} \right) \log \left(\frac{L_i}{O_i} \right) \right). \end{aligned}$$

Here O_i is the opening price.

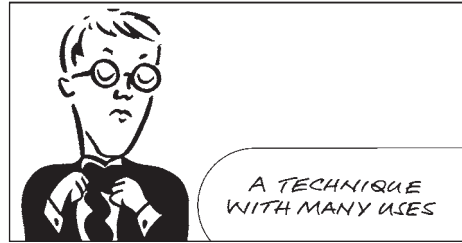
Rogers & Satchell (1991)

Parkinson and Garman & Klass are not independent of the drift. Our final simple volatility estimate is

$$\sigma_{rs}^2 = \frac{1}{n} \sum_{i=1}^n \left(\log \left(\frac{H_i}{C_i} \right) \log \left(\frac{H_i}{O_i} \right) + \log \left(\frac{L_i}{C_i} \right) \log \left(\frac{L_i}{O_i} \right) \right).$$

9.4 MAXIMUM LIKELIHOOD ESTIMATION

A large part of statistical modeling concerns finding model parameters. Two popular ways of doing this are regression and Maximum Likelihood Estimation (MLE). We look at the second method here.



9.4.1 A simple motivating example: taxi numbers

You are attending a math conference. You arrive by train at the city hosting the event. You take a taxi from the train station to the conference venue. The taxi number is 20,922. How many taxis are there in the city?

This is a parameter estimation problem. Getting into a specific taxi is a probabilistic event. Estimating the number of taxis in the city from that event is a question of assumptions and statistical methodology.

The assumptions:

1. Taxi numbers are strictly positive integers.
2. Numbering starts at one.
3. No number is repeated.
4. No number is skipped.

The statistical methodology We will look at the probability of getting into taxi number 20,922 when there are N taxis in the city. This couldn't be simpler, the probability of getting into any specific taxi is

$$\frac{1}{N}.$$

This is shown in Figure 9.4.

Which N maximizes the probability of getting into taxi number 20,922?

$$N = 20,922.$$

This example explains the concept of MLE: *choose parameters that maximize the probability of the outcome actually happening.*

9.4.2 Three hats

You have three hats containing Normally distributed random numbers. One hat's numbers have mean of zero and standard deviation 0.1. This is hat A. Another hat's numbers have

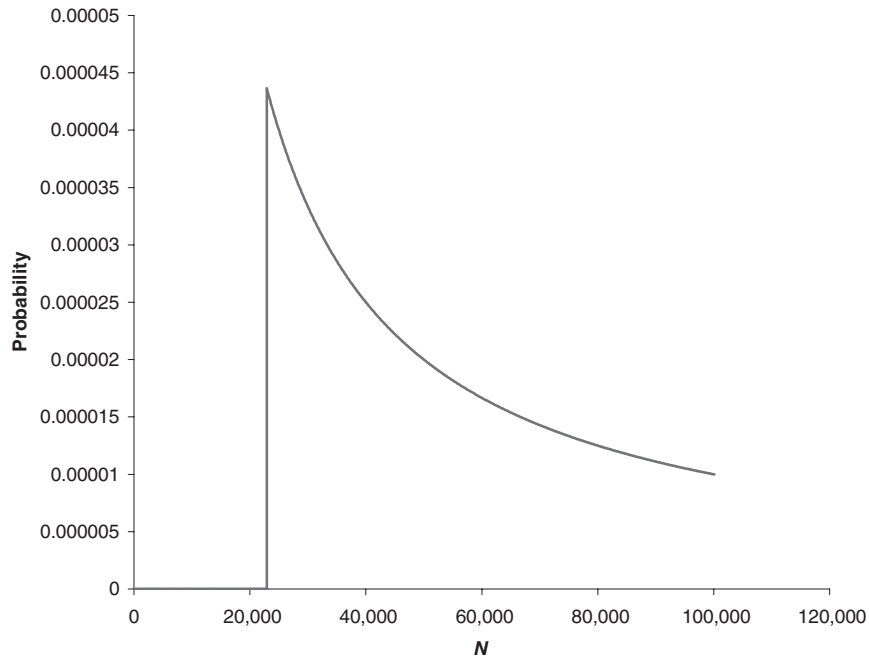


Figure 9.4 Probability of getting into the N th taxi.

mean of zero and standard deviation 1. This is hat B. The final hat's numbers have mean of zero and standard deviation 10. This is hat C.

You don't know which hat is which.

You pick a number out of one hat, it is -2.6 . Which hat do you think it came from?

The 'probability' of picking the number -2.6 from hat A (having a mean of zero and a standard deviation of 0.1) is

$$\frac{1}{\sqrt{2\pi} \cdot 0.1} \exp\left(-\frac{2.6^2}{2 \times 0.1^2}\right) = 6 \times 10^{-147}.$$

Very, very unlikely!

(N.B. The word 'probability' is in inverted commas to emphasize the fact that this is the value of the probability density function, not the actual probability. The probability of picking exactly -2.6 is, of course, zero.)

The 'probability' of picking the number -2.6 from hat B (having a mean of zero and a standard deviation of 1) is

$$\frac{1}{\sqrt{2\pi} \cdot 1} \exp\left(-\frac{2.6^2}{2 \times 1^2}\right) = 0.014,$$

and from hat C (having a mean of zero and a standard deviation of 10)

$$\frac{1}{\sqrt{2\pi} \cdot 10} \exp\left(-\frac{2.6^2}{2 \times 10^2}\right) = 0.039.$$

We would conclude that hat C is the most likely, since it has the highest probability for picking the number -2.6 .

We now pick a second number from the same hat, it is 0.37 . This looks more likely to have come from hat B. We get the following table of probabilities.

Hat	-2.6	0.37	Joint
A	6×10^{-147}	0.004	2×10^{-149}
B	0.014	0.372	0.005
C	0.039	0.040	0.002

The second column represents the probability of drawing the number -2.6 from each of the hats, the third column represents the probability of drawing 0.37 from each of the hats, and the final column is the joint probability, that is, the probability of drawing both numbers from each of the hats.

Using the information about *both* draws, we can see that the most likely hat is now B.

9.4.3 The math behind this: find the standard deviation

You have one hat containing Normally distributed random numbers, with a mean of zero and a standard deviation of σ which is unknown. You draw N numbers ϕ_i from this hat. Estimate σ .

Q. What is the ‘probability’ of drawing ϕ_i from a Normal distribution with mean zero and standard deviation σ ?

A. It is

$$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\phi_i^2}{2\sigma^2}}.$$

Q. What is the ‘probability’ of drawing all of the numbers $\phi_1, \phi_2, \dots, \phi_N$ from independent Normal distributions with mean zero and standard deviation σ ?

A. It is

$$\prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\phi_i^2}{2\sigma^2}}.$$

... choose the σ that maximizes this quantity. This is easy ...

Differentiate with respect to σ (take logarithms first) and set result equal to zero:

$$\frac{d}{d\sigma} \left(-N \log(\sigma) - \frac{1}{2\sigma^2} \sum_{i=1}^N \phi_i^2 \right) = 0.$$

(A multiplicative factor has been ignored here.) I.e.

$$-\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^N \phi_i^2 = 0.$$

Therefore our best guess for σ is given by

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N \phi_i^2.$$

This should be a familiar expression!

9.4.4 Quants' salaries

In Figure 9.5 are the results of a 2004 survey on www.wilmott.com concerning the salaries of quants using the Forum⁴.

This distribution looks vaguely lognormal, with distribution

$$\frac{1}{\sqrt{2\pi}\sigma E} \exp\left(-\frac{(\log E - \log E_0)^2}{2\sigma^2}\right),$$

where E is annual earnings, σ is the standard deviation and E_0 the mean. Using MLE find σ and E_0 .

The MLE solution is shown in Figure 9.6.

The mean $E_0 = \$133,284$, with $\sigma = 0.833$.

If you are a professional 'quant,' how much do you earn?

Last year I earned:

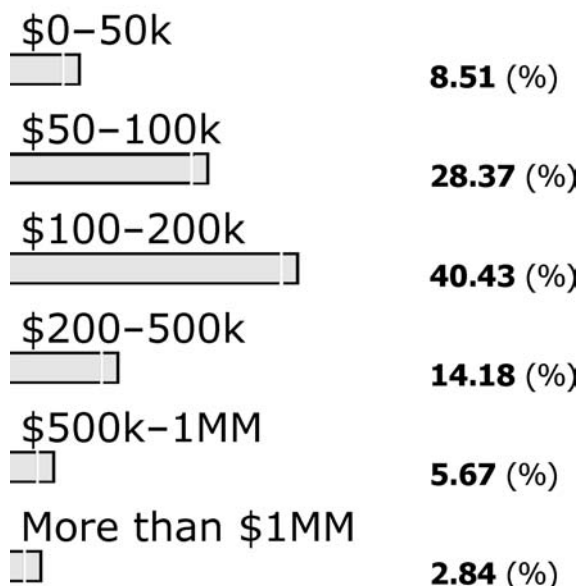


Figure 9.5 Distribution of quants' salaries.

⁴ The respondents were self selecting so the numbers will be biased towards the more forthcoming, that is, less well paid, quant.

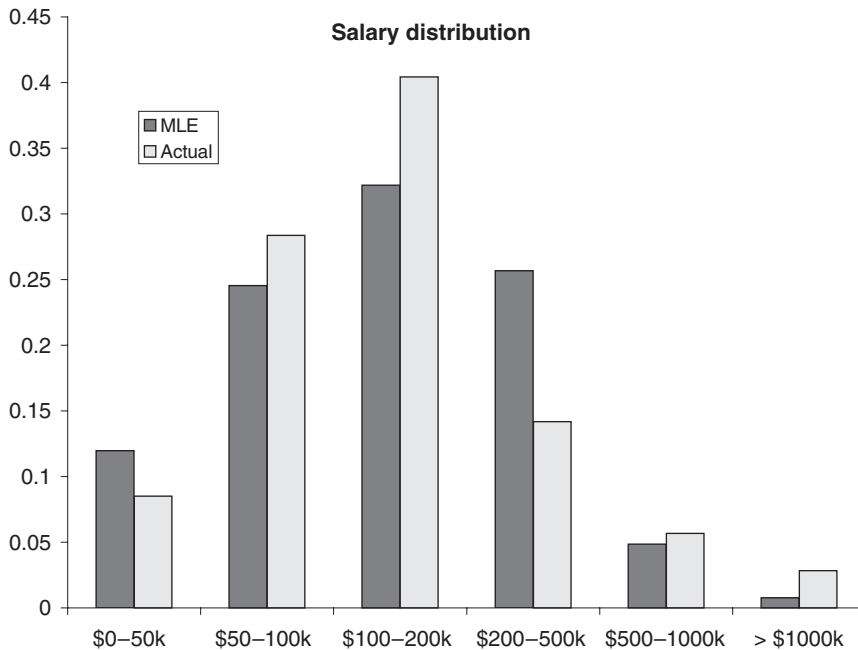


Figure 9.6 Distribution of quants' salaries and MLE fit.

9.5 SKEWS AND SMILES

We have briefly already mentioned skews and smiles, but here is a reminder of what they are about.

For a series of options that all expire at the same date, plot the value of implied volatility against strike. *If* actual volatility were constant, and *if* the Black-Scholes model were correct and *if* people priced options correctly then that plot would be flat, all options would have the same implied volatility. Of course, none of those assumptions is correct, and so there is plenty of scope for that plot to be curved, or skewed. If there is an appreciable slant to that curve, for example if it goes from top left to bottom right, then we have what is called a skew. A negative skew is downward sloping and a positive skew upward sloping. If there is curvature so that the curve has a minimum in the middle then we have a smile. Skews and smiles are the market's way of telling us that either they don't believe in the Black-Scholes model or its assumptions or that they don't care.

'They don't care'? If out-of-the-money puts are expensive, does it matter? Perhaps not if they still cost only pennies and they are the easiest or cheapest way of getting downside protection. To not buy needed downside protection because implied volatility seems a bit high would be a foolish economy. 'Penny wise, pound foolish.'

So smiles and skews may give us information about future volatility or they may give us information about people's expectation of future volatility or they may give us an idea of how desperately they need to hedge. All very useful stuff.

You can speculate on what you think implied volatility might be going to do in the future by buying or selling specific portfolios of vanilla options. If you want to speculate on level

of implied volatility you may buy or sell a straddle, and if you want to speculate on skew then you would consider a risk reversal. Let's see how this works.

9.5.1 Sensitivity of the straddle to skews and smiles

The straddle is a portfolio made up of a long position in a call and a put, both having the same strike and expiration. Because of put-call parity the market will price these two contracts with the same volatility (bid-offer spread aside) and so there is really only one implied volatility to consider. Figure 9.7 shows the Black–Scholes value of this simple portfolio using three different levels of volatility. The strike is 100, the interest rate is 5% and there is one year to expiration. The Black–Scholes value of this portfolio is monotonic in the volatility; increase the volatility and the value rises. Such a portfolio is therefore ideal when it comes to speculating on implied volatility.

9.5.2 Sensitivity of the risk reversal to skews and smiles

The risk reversal is made up of a long call and a short put, the call having a higher strike than the put. Now that there are two strikes to consider we can see that the implied volatility skew is important. Figure 9.8 shows how the value of the risk reversal varies with the skew. In the case of the negative skew we have used a 25% volatility for the put, having strike 80, and a 15% volatility for the call, having a strike of 120. In the positive skew example these quantities are reversed and with the no-skew example both

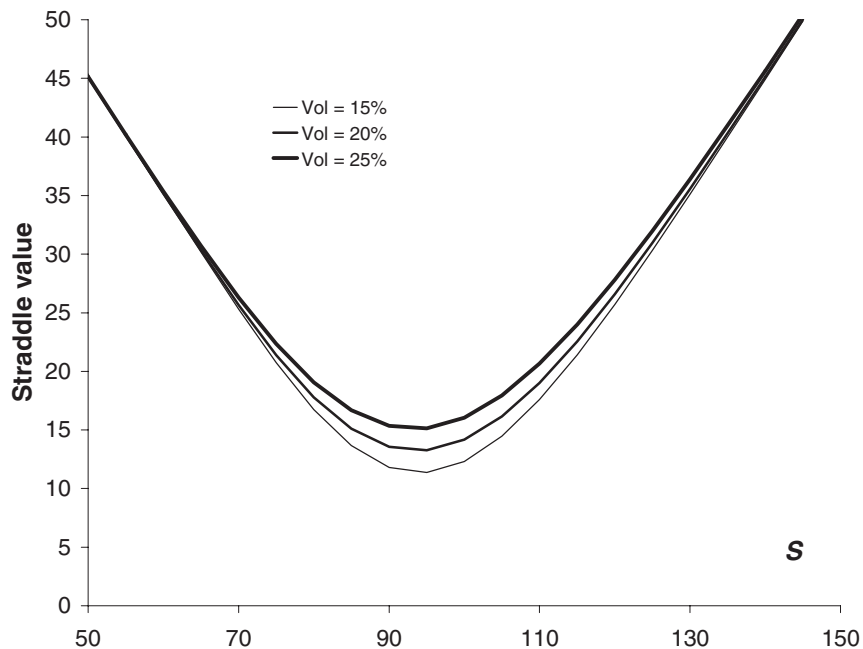


Figure 9.7 How the value of a straddle varies with level of volatility.

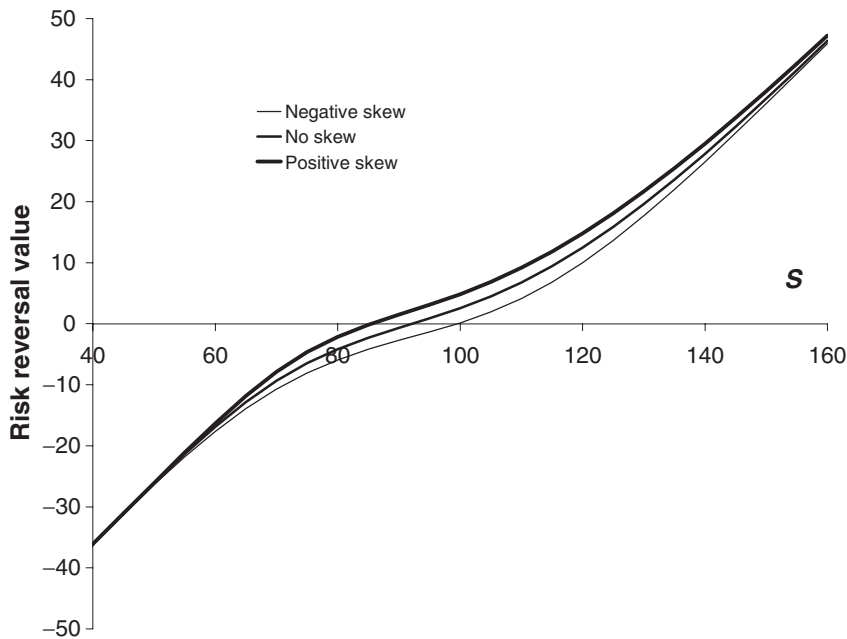


Figure 9.8 How the value of a risk reversal varies with volatility skew.

options have a strike of 20%. The risk reversal's sensitivity to skew makes it perfect for speculation on that skew.⁵

If you want to speculate on the smile what contract would you buy? Simple, you need a payoff with three kinks in it, a butterfly spread.

9.6 DIFFERENT APPROACHES TO MODELING VOLATILITY

For the rest of this chapter I want to provide a brief overview of what is possible in terms of volatility modeling, and to put it all into context. Please see *PWOQF2* for all the details.

9.6.1 To calibrate or not?

Perhaps the biggest question to face is that of how much information should we take from the prices of options in the markets. Given that we can't see what volatility is at any instant, and given that forecasting volatility is not easy (see just a small subset of possibilities above), we might be tempted to use implied volatility as an estimate of future volatility. What we



⁵ At a meeting of The Committee on 15th September 2005 it was unanimously voted that sensitivity to skew be henceforward called 'Xena.' Present were Dominic Connor, Mike Weber and the author.

might say is that implied volatility is the market's best estimate of what volatility will do in the future. In this sense, the implied volatility of a three-month option can be thought of as containing information about actual volatility over the next three months.

Well, yes and no. In some sense, perhaps. But I personally don't believe that the market is that statistically sophisticated. If people want to spend too much buying an option, or charge too little when selling it that is their business, it doesn't mean that their prices are right. It is no different from the pricing of a liter of milk in your corner shop. Has the shopkeeper done some detailed analysis of the utility of individual customers, pricing in the opportunity cost of going to the corner shop versus the cheaper supermarket, allowing for the future price of gas or bus fares, etc.? No, he knows what it costs to buy the milk and he just tries to sell it for as much above that as he possibly can. Out-of-the-money puts are expensive, due to demand, otherwise known as 'fear.' Out-of-the-money calls are cheap, due to supply, otherwise known as 'greed.'

Yes, there is arbitrage. Every business is based on arbitrage, so it is to be expected that arbitrage (or mispricing as theoreticians call it, or earning a living, as sensible people call it) exists. So I wouldn't be too keen on accepting that there is a great deal of information about volatility within option prices.

Anyway, whether to accept prices as containing information is part of the subject of calibration. If you have an option pricing model, should it output theoretical prices that are exactly the same as quoted market prices?

The subject of calibration will crop up a lot from now on.

9.6.2 Deterministic volatility surfaces

The most straightforward modeling approach that leads to theoretical prices that match market prices is to work in the classical Black–Scholes world, with the one exception that actual volatility is not a constant but a function of stock price and time, $\sigma(S, t)$. This is the simplest model that is consistent with the market prices of options. Stock- and time-dependent actual volatility are still entirely consistent with the Black–Scholes partial differential equation, you just have $\sigma(S, t)$ in the coefficient of the gamma term. The big difference from the constant-volatility world is just that we cannot generally find closed-form solutions for option prices anymore.

The way that this idea works is that you have enough freedom in the function $\sigma(S, t)$ to make the theoretical option values (found numerically, say) consistent with all option values in the market, or, totally equivalently, consistent with all implied vols. Now normally you would specify the actual volatility, work out the option prices and so the implied vols. This is the natural forward problem. Not so with this form of calibration. Here we specify the implied vols and ask what actual volatilities are consistent with them. This is what is known to mathematicians as an inverse problem.

Inverse problems are notoriously tricky to solve. One reason is the sensitivity of the result (actual volatilities) to the input (implied volatilities). If you specify actual volatilities and work out implied, then this is a smoothing operation; implied volatilities are averages of actuals. But going the other way does the opposite of smoothing. Make a very small change to implied volatilities and this can have an enormous effect on $\sigma(S, t)$.

If our problem were a diffusion equation from a nice physical problem, this might not matter. Having faith in the diffusion model, finding the parameters (actual volatilities) is a

matter of ‘regularization.’ Unfortunately, we cannot have faith in our model, it is financial not physical, and so the sensitivity to initial input is something we have to live with.

One reason is that common sense says it cannot be right, the financial world is too complicated to obey the rather simplistic $\sigma(S, t)$ model. A second reason is that it is very, very easy to test scientifically. Do the following. Find $\sigma(S, t)$ using option data one day. Then come back a week later, when stock prices and option prices have all changed. Now recalibrate to find $\sigma(S, t)$ again. If the model is right then the first and second calculations will give exactly the same actual volatilities. Do they? No. Never.⁶

9.6.3 Stochastic volatility

Without a doubt it is impossible to accurately forecast future actual volatility so it makes sense to treat that quantity as being random. After all, modeling stock prices as random was the great breakthrough in pricing derivatives based on stocks, so perhaps modeling volatility as random is also a great breakthrough since options are really all about future volatility.

Although stochastic volatility models are commonly used in practice they do suffer from two major problems. First, what model should we choose? What is the volatility of volatility? Given that we can’t even measure volatility at any moment particularly accurately, how on earth can we measure the volatility of that immeasurable volatility?

The second problem concerns hedging. When only the stock price is random we have only one source of randomness. This randomness can be hedged away from an option by using a position in the underlying stock. One source of randomness, one traded quantity for hedging that randomness, end result no randomness at all. Now, when volatility is stochastic we have two sources of randomness, the stock and volatility. But we still only have the one traded quantity to hedge with, the stock. We can’t hedge with volatility to remove volatility risk because that isn’t traded. We can, sort of, get rid of the volatility risk in one option by hedging with another option, an exchange-traded vanilla. End result is a hedged portfolio, but now we have one equation for two unknowns, the value of the original option and the value of the option used for hedging. Oh dear. To get out of that one requires the introduction of a concept known as the market price of volatility risk.

9.6.4 Uncertain parameters

First deterministic volatility, then random volatility. That just about covers the possibilities, no? No. There is also ‘uncertain’ volatility. This is subtly, but importantly, different from random. Random is when you have a probabilistic description, perhaps even a probability density function, for the random variable. Uncertainty is when you have no such concept. Models using uncertainty are therefore far more vague than models using randomness. (And I mean ‘vague’ in a good way.)

⁶ Phlogiston theory: hypothesis regarding combustion. The theory, advanced by J. J. Becher late in the 17th century, and extended and popularized by G. E. Stahl, postulates that in all flammable materials there is present phlogiston, a substance without color, odor, taste, or weight that is given off in burning. ‘Phlogisticated’ substances are those that contain phlogiston and, on being burned, are ‘dephlogisticated.’ The ash of the burned material is held to be the true material. Source: *The Columbia Electronic Encyclopedia*, 6th ed. 2004. Many financial theories seem very phlogiston-like to me. It is time for some debunking!

The simplest such model allows volatility to lie within a range. But there is no mention made of how likely the volatility is to be at any point in that range, it is a genuine model of uncertainty not of randomness. Given a range of possible values for volatility, we find that there is a range of possible values for an option. Furthermore, long and short positions take different values, and the whole business of pricing derivatives becomes non linear.

9.6.5 Static hedging

Hedging is used to reduce or, if possible, to eliminate risk. In options theory and practice we dynamically delta hedge to eliminate the stock-price risk in an option. Such theory we have seen several times in this book. But there are other, perhaps more obscure, risks that we are only just starting to worry about here. One of those risks is caused by volatility. The plain and simple fact is that we don't know what volatility is going to be in the future therefore when we price an option we will be exposed to the input volatility being wrong, we therefore have volatility risk.

In practice, instead of just dynamically hedging with vanilla options, we also statically hedge. To see how this works, imagine that we want to price and hedge some exotic contract. Suppose that the payoff from that exotic contract can somehow be closely matched by the payoff from a portfolio of vanillas. If it can, then you hedge with those vanillas leaving only a small residual payoff which needs delta hedging. The original contract may have been very difficult to delta hedge, and so very exposed to your volatility input. But after statically hedging with the vanillas the delta-hedging task is much easier. And, of course, you know exactly what the portfolio is worth (the contracts are exchange traded) so you know almost exactly what the exotic should be worth. In practice, you won't be able to get a perfect match between the exotic and the vanillas because if you could then the exotic wasn't exotic at all.

9.6.6 Stochastic volatility and mean-variance analysis

As mentioned above, one of the drawbacks with stochastic volatility models is that when you build up the governing equation you assume that you can dynamically hedge with options. And the equation you end up with contains the market price of volatility risk, a wonderful concept in theory but rather too unreliable for practice. There are ways of tidying this all up without these two problems, and that is to accept that not all risk can be dynamically hedged away, and then to assign a value to that risk.

9.6.7 Asymptotic analysis of volatility

When deciding which stochastic volatility model to choose we often face the dilemma of whether to pick an easy-to-compute model that might not be so accurate, or a slower but more scientifically precise model. Practitioners almost always go for fast and inaccurate. However, by exploiting a little known applied math concept known as asymptotic analysis⁷ you can actually get the best of both worlds. Asymptotic analysis

⁷ Little known in finance, at least. To applied mathematicians asymptotic analysis is a very commonly used tool in their toolbox. Prior to succumbing to the lure of finance almost every mathematical model I worked on used asymptotic analysis at some stage, mainly because they would otherwise have been far too complicated to solve or understand.

is all about finding approximate solutions to differential equations, models, whatever, by exploiting the relative largeness or smallness of a parameter in the model. If some terms in a complicated equation are multiplied by a small number then perhaps those terms don't matter so we can ignore them, leaving us with a simpler equation that we may be able to solve. Of course, it is nowhere near as simple as that, but it gives you the flavor of the technique.

9.7 THE CHOICES OF VOLATILITY MODELS

Model	Math	Popularity
Constant vol	$\sigma = \text{constant}$, Black–Scholes formulæ	Very, especially for vanillas
Deterministic vol	$\sigma(S, t)$, Black–Scholes pde	Very, for exotics
Stochastic vol	$d\sigma = \dots$, higher dimensions, transforms	Very, for exotics
Jump diffusion	Poisson processes, jumps in stock and/or vol.	Increasing
Uncertain vol	$\sigma^- \leq \sigma \leq \sigma^+$, non-linear pde	Not at all, unfortunately
Stoch vol and mean-variance	$d\sigma = \dots$, higher dimensions, non-linearity	Not at all, unfortunately

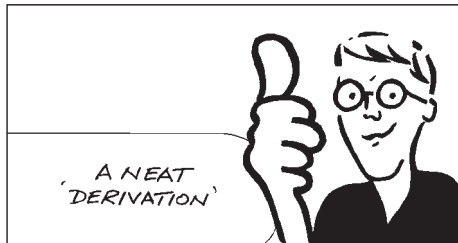
9.8 SUMMARY

Derivatives are all about volatility. You can't price or hedge derivatives without a decent model for volatility. And if your volatility model is better than the market's you could make money by speculating on volatility. We will see this in the next chapter. One point to watch out for is how much information we back out from vanilla option prices into our volatility model. Never forget that volatility is a property of the stock and would still be present even if there weren't any derivatives! Indeed, how on earth can markets possibly have perfect knowledge of the future, and the future volatility of the stock? Of course they can't. Option prices may be governed to some extent by what people expect to happen in the future in some rational sense, but they are also governed by fear and greed as interpreted by option prices through simple supply and demand. Option prices, and hence volatility, will rise when people panic and rush to buy those OTM puts, regardless of whether this panic is rational. And if you do have to pay 10 cents for an option that may only be worth 7 cents, well so what? It's still pennies after all. However, translate those pennies into implied volatility numbers and suddenly it looks like the market is expecting volatility to rise.

FURTHER READING

- Natenberg (1994), of course.
- Shu Zhang (2003) look at data for SP500 index option volatility and compare with realized volatility.

- And for something more (much more) quanty, see Rebonato (2004).
- Javaheri (2005) on volatility arbitrage is a must read.
- A survey of stochastic volatility models, their calibration and how very differently they price the same contracts is given in Schoutens, *et al.* (2004).
- All of the models discussed above are described in great detail in *PWOQF2*.



APPENDIX: HOW TO DERIVE BS PDE, MINIMUM FUSS

Here is how to derive the Black–Scholes equation, with the minimum fuss and the minimum (explicit) assumptions. This derivation also shows the special, almost arbitrary role, played by volatility.

Working with any Brownian-motion type model for stock price dynamics will give you a linear diffusion equation for pretty much everything, including option values. If $V(S, t)$ is the option's value then the general linear diffusion equation (and homogeneous, since the value of an empty portfolio should be zero) is

$$\frac{\partial V}{\partial t} + a \frac{\partial^2 V}{\partial S^2} + b \frac{\partial V}{\partial S} + cV = 0$$

where a , b and c are, for the moment, arbitrary.

A solution of this equation must be cash, i.e. $e^{-r(T-t)}$. Plug this function into the partial differential equation to find that

$$c = -r.$$

Another solution is the stock, S . Plug this function in and you will find

$$b = rS.$$

This gives you the risk-neutral pricing equation

$$\frac{\partial V}{\partial t} + a \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0.$$

The only remaining 'real' or 'arbitrary' parameter left to determine or fudge/calibrate is a . And we know (from what we've done before) that this is related to volatility. So, this shows us a couple of things:

- Basic considerations pin down the coefficients of most of the terms in the pricing equation, and are clearly risk neutral as well.
- To determine or model the diffusion coefficient you will need to go to greater lengths. Or more simply, just use this function as the only one you can arguably calibrate.

EXERCISES

1. Using real, daily data, for several stocks, plot a time series of volatility using several models.
 - (a) Divide the data into yearly intervals and estimate volatility during each year.
 - (b) Use a fixed-length, moving window.
 - (c) Use an exponentially weighted estimate.
2. Collect real option data from the *Wall Street Journal*, the *Financial Times* or elsewhere, calculate implied volatilities and plot them against expiration, against strike, and, in a three-dimensional plot, against strike and expiration.

